Dynamical Localization of Quantum Walks on the hexagonal lattice in the regime of strong disorder

Cergy 2025 - Quantissima sur Oise

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Quantum Walks on the hexagonal lattice

Intuition: Quantum particle moving in discrete time on discrete grid Γ.
State of quantum particle with internal degree of freedom s is given by $\psi \in l^2(\Gamma) \otimes \mathbb{C}^s$.

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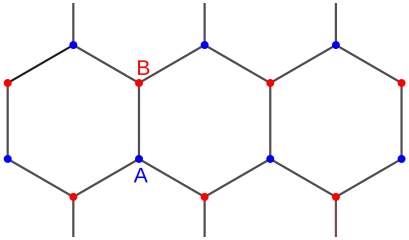
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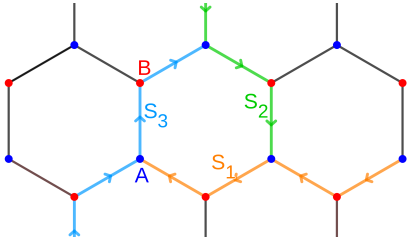
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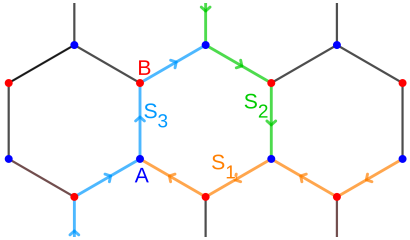
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$\Longrightarrow$ Quantum Walk: $U = \mathcal{SC}$



Transport Properties

Translation-invariant case:

Fourier Transform: $\widehat{U} = \mathcal{F} U \mathcal{F}^{-1}$ is multiplication operator on $L^2(\mathbb{T}^2, \mathbb{C}^6)$

$\rightsquigarrow \sigma_{\text{cont}}(U) = \varnothing \iff k \in \mathbb{T}^2 \mapsto \text{tr}\Big(\widehat{U}(k)\Big) = \text{const.}$

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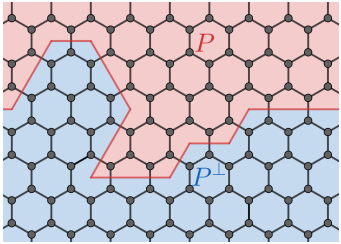
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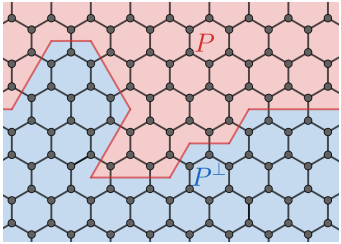
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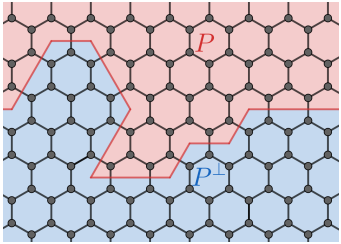
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$\text{ind}(U) \neq 0 \Rightarrow$ criteria on $\sigma(U), \sigma_{\text{ac}}(U) = \mathbb{S}^1$, e.g. $|(C_x)_{2,1}|, |(C_x)_{1,2}|, |(C_x)_{3,3}| \to 1$ ($|x| \to \infty$ on boundary), then $\sigma(U) = \mathbb{S}^1$



Dynamical Localization

Equip every site $x \in \Gamma$ with i.i.d. RV ω_x uniformly distributed on $[0, 2\pi]$
$\Longrightarrow$ define $D_\omega(|x\rangle \otimes e_i) = e^{i\omega_x} |x\rangle \otimes e_i$ and $U_\omega = D_\omega U$

Definition

Dynamical localization: $\exists\, g > 0$ such that $\forall\, x, y \in \Gamma$:

$$\mathbb{E}_\omega \Big(\sup_{n \in \mathbb{Z}} |\langle x| U_\omega^n |y\rangle| \Big) \leqslant c\, e^{-g|x-y|}$$

i.e. probability to move from site y to x decreases exponentially in distance $|x - y|$, independently of how many steps QW may take

Dyn. Loc. implies spectral localization, i.e. U_ω has pure point spectrum a.s.

Dynamical Localization

Fixing a root $|0\rangle$ of the lattice, we define the position operator $X\,|x\rangle = |x|\;|x\rangle$ and compute the expected average spread at time n:

$$\mathbb{E}_\omega\left(\|X^{\frac{p}{2}}\,U_\omega^n\;|0\rangle\,\|^2\right) = \sum_{|x\rangle} |x|^p\;\mathbb{E}_\omega\Big(|\langle x\,|\,U_\omega^n\,|0\rangle|^2\Big) \leqslant \sum_n n^p\;C\;|S_n|\;e^{-gn}$$

Here: $S_n = \{|x\rangle \text{ s.t. } |x| = n\}$.
The RHS is finite iff $|S_n| = o(e^{\gamma n})$ for some $\gamma < g$.

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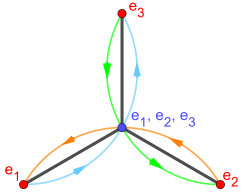
Here: $S_n = \{|x\rangle \text{ s.t. } |x| = n\}$.
The RHS is finite iff $|S_n| = o(e^{\gamma n})$ for some $\gamma < g$.
Fractional moments criterion: for some $s \in (0,1)$ and $1-\epsilon < |z| < 1$

$$\mathbb{E}_\omega\Big(|\,\langle x|\,(U_\omega - z)^{-1}\,|y\rangle\,|^s\Big) \leqslant c\,e^{-g|x-y|}$$

E. Hamza, A. Joye, G. Stolz (2009): Frac. mom. crit $\Longrightarrow$ dyn. loc.

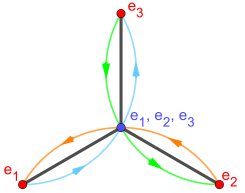
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Theorem

For any $s \in (0, \frac{1}{3})$ there exists $\varphi, c, g > 0$ such that for any $\{C_x\}_{x\in\Gamma} \subset U(3)$ with $\sup_x \|C_x - C_0\| \leqslant \varphi$ and all $x, y \in \Gamma$, $z \in \mathbb{C}$ with $|z| < 1$:

$$\mathbb{E}_\omega \left(\left| \langle x \otimes e_i | \left(U_\omega - z \right)^{-1} | y \otimes e_j \rangle \right|^s \right) \leqslant c\, e^{-g|x-y|}.$$

Proof: Finite volume method

Finite volume method

Restrict QW U_ω to box Λ_L of size $L \in \mathbb{N}^2$

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Iterative step:

$\exists\, q < 1$ s.t. for $s < \frac{1}{3}$, L big enough $\exists\, \varphi > 0$:

$$\mathbb{E}_\omega\left(\left|\langle x^{(i)}|\,(U_\omega - z)^{-1}\,|y^{(j)}\rangle\right|^s\right) \leqslant q \max_{v^{(k)}\in\partial(\Lambda_L+x)} \mathbb{E}_\omega\left(\left|\langle v^{(k)}|\,(U_\omega - z)^{-1}\,|y^{(j)}\rangle\right|^s\right)$$

for all $\sup_x \|C_x - C_0\| \leqslant \varphi$ and y outside of box $\Lambda_L + x$ (N.B: $\left|x^{(i)}\right\rangle = |x\rangle \otimes e_i$)

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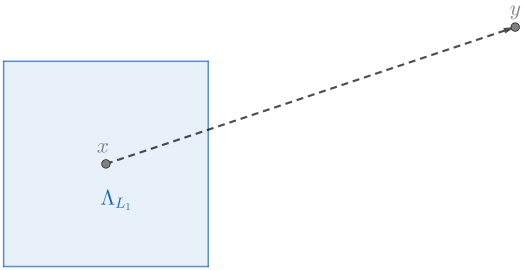
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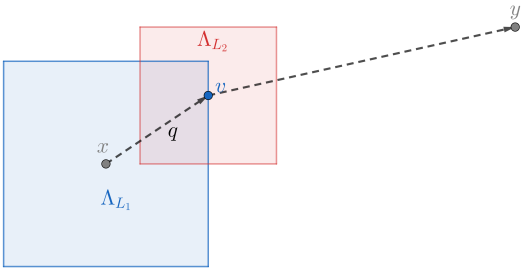
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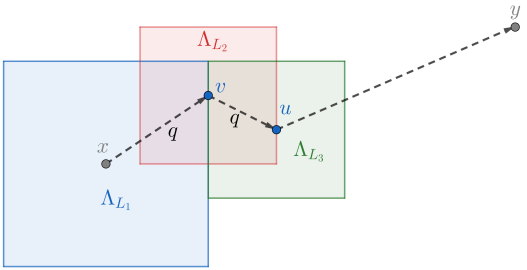
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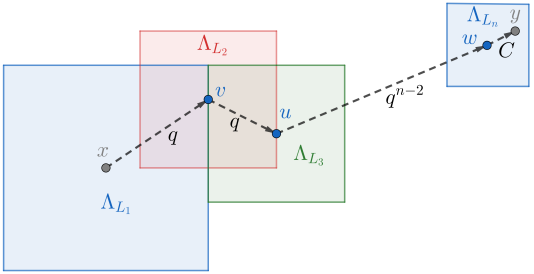
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LHS $\leqslant C\, q^n$

n depends on $|x - y|$

Correlated phases

Recall $D_\omega(|x\rangle \otimes e_i) = e^{i\omega_x}\ |x\rangle \otimes e_i$, we write D_ω in the ONB $\left(|x\rangle \otimes e_i\right)_{x\in\Gamma, i=1,2,3}$:

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Problem: "Fractional Moment Criterion $\Rightarrow$ Dynamical Localization" requires independent phases

Thank you for your attention!